



Maharashtra Education Society's
Institute of Management and Career Courses (IMCC), Pune
AUTONOMOUS
Academic Year 2026-27



DSA Practice Questions

- 1. Calculate Array Sum:** Write a program to accept N integers into an array and calculate and display the sum of all the elements.
- 2. Find the smallest and Largest Element:** Write a program to accept N integers into an array and find and display the largest element, second largest element, smallest element, second smallest element present in the array.
- 3. Count Even and Odd Numbers:** Write a program to accept N integers into an array and count and display the number of even and odd elements present in the array.
- 4. Search an Element:** Write a program to accept N integers into an array and search for a given number. Display an appropriate message indicating whether the number is present in the array or not and also display its position.
- 5. Reverse the Array:** Write a program to accept N integers into an array and display the elements in reverse order without changing the original array.
- 6. Remove Duplicate Elements:** Write a program to accept N integers into an array and create a new array containing only the unique elements, removing all duplicate values.
- 7. Move Zeros to the End:** Write a program to accept N integers into an array and rearrange the elements so that all 0 values are moved to the end while maintaining the relative order of the non-zero elements.

8. Print following patterns

```

      *
      *
      *
*   *   *   *   *   *   *
      *
      *
      *
      *
  
```

```

*   *   *   *   *
*           *
*           *
*           *
*           *
*   *   *   *   *
  
```

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
 * * * * *
  * * * *
   * * *
    * *
     *
  
```

```

      1
     1 2 1
    1 2 3 2 1
   1 2 3 4 3 2 1
  1 2 3 4 5 4 3 2 1
  
```

```

1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
0 1 0 1 0
1 0 1 0 1
  
```

9. Count: Write a program to count vowels, consonants, digits and special characters in a given string.

10. Count Substrings: Find all occurrences of substring in a given string.

11. Reverse Words: Write a program to accept a sentence and display by reversing words in a sentence.

Ex. "the sky is blue" => "eht yks si eulb"

12. Find longest and shortest words: Write a program to accept a sentence and find the longest and shortest word in a sentence.